Content-Invariant Dual Learning for Change Detection in Remote Sensing Images

Bo Fang, Gang Chen, Guichong Ouyang, Jifa Chen, Rong Kou, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— With the introduction of artificial intelligence (AI), current deep learning-driven change detection methods generally regard changes as a type of specific land cover information and try to detect them simply by using existing semantic labeling models. In practice, the changes that occur on different land cover categories may appear completely different for remote sensing images, making it difficult to detect multiple categories of changes with an end-to-end model. In addition, training such networks requires a large amount of prelabeled references, which are labor-intensive and time-consuming. Motivated by this observation, we integrate dual learning algorithm and disentangled representation theory to develop a novel approach, named content-invariant dual learning (CiDL), for either supervised or unsupervised change detection in remote sensing images. In our framework, two opposite Y-shaped networks, each of which consists of two encoders and one decoder, are introduced to translate bitemporal images from their original domains to each others, where their intrinsic content features are retained, while their style features are consistent. By training our hybrid framework, even without references, this method learns a category-wise cross-domain translation to suppress the discrepancies in paired unchanged regions, and meanwhile highlight those in paired changed regions. The experimental results on two typical change detection data sets and the comparison with other state-of-the-art deep learning-driven methods verify the effectiveness and competitiveness of our proposed CiDL.**

***Index Terms*— Change detection, disentangled representation, dual learning, generative adversarial networks (GAN), remote sensing.**

NOMENCLATURE

C	Content feature.
$\mathcal{C}$	Content space.
CD	Change detection.
CM	Reference change map.
D	Domain discriminator.
De	Joint decoder.
En^C	Content encoder.
En^S	Style encoder.
I	Image or image patch.
$\mathcal{L}$	Loss function.
P	Predicted change map.
RS	Remote sensing.
S	Style feature.
$\mathcal{S}$	Style space.
T	Time or domain.
λ	Relative importance.

Manuscript received July 6, 2020; revised November 14, 2020 and February 17, 2021; accepted March 4, 2021. Date of publication March 26, 2021; date of current version December 13, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 41674015, Grant 41925007, and Grant U1711266, and in part by the Scientific Research Project of Hubei Province under Grant 1232039. *(Corresponding author: Gang Chen.)*

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Digital Object Identifier 10.1109/TGRS.2021.3064501

I. INTRODUCTION

CHANGE detection, one of the most fundamental and vital tasks in RS, mainly refers to the process of analyzing two or more RS images acquired over the same geographic area but at different times, and then identifying the changed regions [1], [2]. It plays an important role in extensive applications, such as land cover change mapping [3], environmental monitoring [4], vegetation studying [5], urban planning [6], and disaster assessment [7]. With the rapid development of Earth observation technology, an increasing number of RS images with high spatial and spectral resolutions are available, providing considerable information but more redundancy and noise. This leads to new opportunities but challenges for CD tasks [8].

For the last decade, the development of deep learning technologies has displayed powerful capabilities in terms of feature extraction and expression [9]–[11], providing new ideas for most image processing applications. Various classic neural networks have been applied in RS specifically for CD tasks and achieved impressive performance. They mainly include recurrent neural networks (RNNs) with long short-term memory (LSTM) [12], basic convolutional neural networks (CNNs) [13], fully convolutional networks (FCNs) [14], and generative adversarial networks (GANs) [15]. These models can exploit the implicit features from paired RS images, which will significantly improve the CD accuracy and effectively avoid the negative influences of redundancy and noise.

Analogous to the conventional CD algorithms, current deep learning-driven methods can be divided into three primary categories: original image-based, image algebra-based, and image transformation-based.

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Technically, most deep learning-driven methods are mainly based on the original images. In these approaches, the detection models generally operate the two original images or their concatenation, and then produce the predicted change maps. Gong *et al.* [16] trained a deep belief network to directly classify the changed and unchanged regions using synthetic aperture radar (SAR) images. Lyu *et al.* [17] applied a single LSTM network to learn the transferable change rules for end-to-end land cover CD. Wang *et al.* [18] proposed an end-to-end 2-D CNN for CD using hyperspectral images. This type of method is advantageous because of its simplicity and directness. Nevertheless, changes in RS images are agnostic to any particular characteristic, and they do not belong to each category of land cover. Therefore, in some scenarios, detecting diverse types of changes just with an end-to-end semantic labeling model is not desirable enough.

Image algebra-based methods concentrate on the pixel-level features and generally begin with the calculation of the image difference [19] and image ratio [20], or the analysis of change vector (CVA) [21], [22], and then predict the changes based on the difference image or its concatenation with the original images. Gao *et al.* [23] first filtered the paired images with Gabor wavelets and fuzzy c-means to select the interested pixels, and then separated the changed and unchanged pixels using a principal component analysis network (PCANet). Gong *et al.* [24] took initial difference images as prior knowledge for sampling the training data and applied GANs to generate the final change maps. Yang *et al.* [25] took difference images as auxiliary data and proposed a transferred deep learning-based framework for CD without any reference label. This type of method can effectively utilize the structure and correlation information between bitemporal images, thereby reducing the negative influences of redundancy and noise. Nevertheless, the difference image itself cannot maintain the abundant information and the complete intrinsic characteristics of land covers, which are involved in the original images. This may induce more blur and uncertainty for CD tasks to some extent.

Image transformation-based methods focus on domain attributes and generally transform the bitemporal images into a same domain space at first, and then predict changes by comparing or analyzing both the translated images. Zhan *et al.* [26] proposed a deep Siamese convolutional network (DSCN), derived from the Siamese networks, to translated paired images into a same domain space and detect changed regions by a contrastive loss. Liu *et al.* [27] expanded DSCN by proposing an unsupervised symmetric convolutional coupling network (SCCN) to identify changes between optical and SAR images. Chen *et al.* [28] introduced a deep Siamese convolutional multiple-layer RNN for CD in homogeneous and heterogeneous images. Niu *et al.* [29] adopted a conditional GAN (cGAN) to transform heterogeneous images into a consistent representation space, and then predict changes using an approximation network. Inspired by the Siamese mechanism and skip connection technique, Hou *et al.* [30] first proposed a W-Net to detect changes using optical RS images, and then integrated a W-Net and a GAN to build a CDGAN for improving the CD accuracy. This type of method can effectively suppress the differences in the paired unchanged regions and highlight those in the paired changed regions, facilitating the reduction of negative impacts from domain discrepancies between bitemporal images. However, current deep learning-driven image transformation-based methods still depend on sufficient pixel-level reference change maps for all data sets, which are extremely labor-intensive and time-consuming.

For registered bitemporal RS images, in an ideal situation, the intrinsic land cover information of the paired unchanged regions should visually remain consistent and distinctive discrepancies should exist only in the changed regions. In reality, temporal spectral variability tends to exist in all bands throughout the paired images, as a result of different imaging sensors, capture times, and illumination and atmospheric conditions. This makes land cover samples distribute discretely and irregularly in the chromatic space, no matter changed or unchanged ones they are, as illustrated in Fig. 1(a). At this point, the distance of the unchanged samples in the paired images may be as large as that of the changed ones. Therefore, it is technically difficult to simultaneously identify the changed and unchanged regions directly from the original paired images by using simple end-to-end deep learning models. To address this problem, in our previous research [31], a dual learning-based Siamese framework (DLSF) that integrates ideas of dual learning and Siamese network was introduced for supervised CD using bitemporal RS images. It translates two paired images from their original domains to each others, where the unchanged samples distribute closely, while the changed ones appear entirely diverse. At this point, it is easy to separate the changed and unchanged regions by directly comparing the paired original and translated images or using a simple detector, as illustrated by the red dotted circle in Fig. 1(b). Nevertheless, DLSF remains an end-to-end model and is unable to consider multiple categories of changes, leading to certain limitations in terms of precision and generalization. Similar to other end-to-end CD models, DLSF also yields unsatisfactory performance for complex changes.

According to the above analyses, in this article, we propose a novel image transformation-based framework, named content-invariant dual learning (CiDL), for CD using RS images. This framework involves two opposite Y-shaped networks (Y-Nets) and two domain discriminators. Each Y-Net contains two encoders and one decoder, which aim to extract content and style features from the images and reconstruct images by combining the extracted content and style features. With the diverse content features and consistent style features, the two Y-Nets can simultaneously translate paired RS images from their original domains to each others while retaining the intrinsic land cover information. Meanwhile, the two domain discriminators aim to evaluate the domain consistencies of paired original and translated images. By training this framework, our method learns a category-wise cross-domain translation, which is able to effectively suppress the differences in paired unchanged regions and highlight those in paired changed regions. The primary contributions of our research can be summarized as follows.

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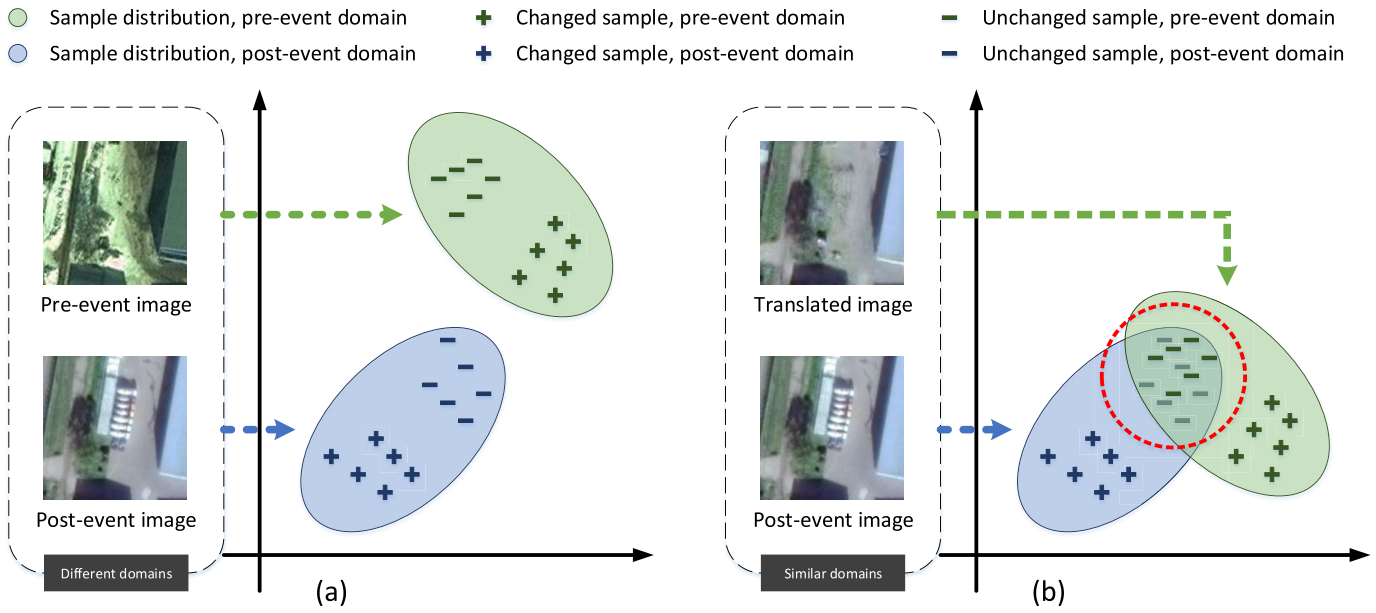


Fig. 1. Illustration of our image transformation-based strategy for change detection in remote sensing images. (a) Sample distributions of the pre-event and postevent images. (b) Sample distributions of the translated and postevent images.

1) We propose a novel image transformation-based method for either supervised or unsupervised CD tasks using RS images. To the best of our knowledge, this is the first time simultaneously applying ideas of dual learning and disentangled representation in CD.
2) We introduce two novel objectives consist of several constraints to incentivize our hybrid framework, respectively, for the unsupervised and supervised learnings. During the cross-domain translation, the latent reconstruction losses are highly emphasized.

The remainder of this article is organized as follows. Related works about our research are briefly described in Section II. The theory and relevant implementation of our proposed CiDL are introduced in Section III. The experimental results on two typical data sets are presented in Section IV. Relative analyses to verify the effectiveness and competitiveness of our models are provided in Section V. Finally, our conclusion is summarized in Section VI.

II. Related Work

A. Generative Adversarial Networks

In recent years, GANs have already witnessed rapid progress in image generation and domain adaptation. Existing adversarial learning models are generally derived from cGANs [32], which apply GANs in conditional settings. Specifically, for image-to-image translation, the primary goal is to learn a mapping model $G : (x, z) \rightarrow y$, from the input image x and random noise vector z to the translated image y. The generator G is trained to produce outputs that cannot be distinguished from "real" reference images by a discriminator D, which is trained to detect generated outputs as "fake" images as well as possible. In this two-player min-max game, the cGANs are trained to learn how to effectively perform the mapping from input images to translated images, and this process is incentivized by an adversarial loss, expressed as

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{x,y}\big[\log D(x, y)\big] + \mathbb{E}_{x,z}\big[\log(1 - D(x, G(x, z)))\big] \quad (1)$$

where G tries to minimize this objective against an adversarial D that tries to maximize it, and the solution is expressed as

$$G^* = \arg\min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D). \quad (2)$$

This type of network has been widely applied in many applications such as image translation [33], image colorization [34], super resolution [35], domain adaptation [36], and style transfer [37]–[39]. Due to the specific mechanism, GANs are practically appropriate to be the backbone of image transformation-based CD methods.

B. Dual Learning

The dual learning algorithm involves making a loop translation between two types of data by setting two bidirectional task mappers, which are a primal mapper $f : x \rightarrow y$ and a dual mapper $g : y \rightarrow x$. The input signals are mapped forward $y = f(x)$ and backward $\tilde{x} = g(y)$, then the feedback signals are reconstructed. With the deviation of the input and feedback signals $x - g(f(x))$, these two mappers represent a primal-dual pair, which try to teach each other through a reinforcement learning process. By minimizing the reconstruction loss, the primal and dual models can be updated alternately to better achieve translations via a policy gradient algorithm, expressed as

$$\begin{cases} f = f + \eta\nabla f, g = g + \eta\nabla g, & \text{if } x - g(f(x)) > t \\ f = f - \eta\nabla f, g = g - \eta\nabla g, & \text{if } x - g(f(x)) < t \end{cases} \quad (3)$$

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where η is the learning rate, and t is the threshold. ∇f and ∇g are the gradients of the two models.

This algorithm was first introduced in natural language processing [40]. Thereafter, some researchers found that it worked equally well for unsupervised image-to-image translation, such as in DualGAN [41], DiscoGAN [42], and CycleGAN [43]. By considering the implicit relationship in the bitemporal images, dual learning achieves bidirectional domain transformation between the pre-event and postevent domains. For CD tasks, this symmetric structure is more reliable and robust than one-sided domain adaptation strategies.

C. Disentangled Representation

The disentangled representation theory attempts to model the factors of data variations, in order to simulate the human visual system. Previous work generally makes use of labeled data to factorize representations to category-dependent and category-independent attributes [44], [45]. Thereafter, specific strategies have been applied to explore unsupervised settings. InfoGAN [46] achieved disentanglement by maximizing the mutual information between latent variables and data variation. DrNet [47] separated time-varying and time-independent attributes with an adversarial loss.

Following the human visual system, images are supposed to be composed of content and style attributes, where the former refers to the underlying spatial structure, and the latter refers to the rendering of the structure. In computer vision, learning the disentangled representation of images can effectively separate the content and style attributes, facilitating the understanding of certain implicit information of images. DRIT [48] introduced a disentangled representation framework with a cross-cycle consistency loss to achieve unsupervised image-to-image translation between unpaired images. MUNIT [49] proposed a multimodel unsupervised image-to-image translation framework, to translate an image to another domain by combining its content code with a random style code sampled from the style space of the target domain. Disentangled representation facilitates specific tasks that focus on a single class of features by separating the content and style features of images. Since CD mainly concerns the changes in paired content features, the application of this theory is able to alleviate the negative influences from the style discrepancies.

III. Methodology

The formulation of our CiDL for CD tasks using RS images is first presented in this section. Thereafter, our new objectives are introduced and interpreted in detail. Finally, relevant implementation regarding the network architectures, training details, and predicting details is depicted.

A. Problem Formulation

Let I_1 and I_2 denote registered bitemporal images acquired over the same geographic area but at different times T_1 and T_2, respectively. According to the disentangled representation theory, these two paired images can be decomposed into the latent spaces: a shared domain-invariant content space $\mathcal{C}$ and two separate domain-specific style spaces $\{\mathcal{S}_1, \mathcal{S}_2\}$, defined as

$$\{C_1, S_1\} = \left\{\mathrm{En}_1^C(I_1), \mathrm{En}_1^S(I_1)\right\} C_1 \in \mathcal{C}, \quad S_1 \in \mathcal{S}_1 \tag{4}$$

$$\{C_2, S_2\} = \left\{\mathrm{En}_2^C(I_2), \mathrm{En}_2^S(I_2)\right\} C_2 \in \mathcal{C}, \quad S_2 \in \mathcal{S}_2 \tag{5}$$

where $\{\mathrm{En}_1^C, \mathrm{En}_1^S\}$ and $\{\mathrm{En}_2^C, \mathrm{En}_2^S\}$ are the content and style encoders for the two different domains, respectively. They aim to extract the content and style features from the input images, by translating them from the visual space into the latent spaces. In practice, these two types of features can be reconstitute into the original images, defined as

$$I_{1\to 1} = \mathrm{De}_1(C_1, S_1) \tag{6}$$

$$I_{2\to 2} = \mathrm{De}_2(C_2, S_2) \tag{7}$$

where De_1 and De_2 are the joint decoders for the two different domains, respectively. They are intended for generating new images with the given content and style features, by translating them from the latent spaces to the visual space.

With the aforementioned two types of models, our proposed CiDL is designed and illustrated in Fig. 2. This framework comprises two dual data streams: from domain T_1 to T_2, and from domain T_2 to T_1. Each stream is driven by a Y-Net, where the two encoders and one decoder form an architecture shaped like a "Y." With these two streams, the bitemporal RS images are translated to each others domains, expressed as

$$I_{1\to 2} = \mathrm{De}_1\left(\mathrm{En}_1^C(I_1), \mathrm{En}_2^S(I_2)\right) \tag{8}$$

$$I_{2\to 1} = \mathrm{De}_2\left(\mathrm{En}_2^C(I_2), \mathrm{En}_1^S(I_1)\right) \tag{9}$$

where $I_{1\to 2}$ denotes the translated image that has similar content distribution as I_1 and similar style appearance as I_2, and $I_{2\to 1}$ denotes the one that has similar content distribution as I_2 and similar style appearance as I_1. Remarkably, the two translated images can also be translated back to their original domains by using the two Y-Nets, expressed as

$$I_{1\to 2\to 1} = \mathrm{De}_2\left(\mathrm{En}_2^C(I_{1\to 2}), \mathrm{En}_1^S(I_{2\to 1})\right) \tag{10}$$

$$I_{2\to 1\to 2} = \mathrm{De}_1\left(\mathrm{En}_1^C(I_{2\to 1}), \mathrm{En}_2^S(I_{1\to 2})\right) \tag{11}$$

where $I_{1\to 2\to 1}$ denotes the reconstructed image that has similar visible information as I_1, and $I_{2\to 1\to 2}$ denotes the one that has similar visible information as I_2.

To enable effective learning of the two Y-Nets, two domain discriminators D_1 and D_2 are introduced to evaluate the domain consistency of the registered original and translated images. As the adversaries of the encoders and decoders, these two discriminators try to identify the original images as real images, while identifying the translated images as fake ones, expressed as

$$\begin{cases} D_1(I_1) = \mathrm{TRUE} \\ D_1(I_{2\to 1}) = \mathrm{FALSE} \end{cases} \tag{12}$$

$$\begin{cases} D_2(I_2) = \mathrm{TRUE} \\ D_2(I_{1\to 2}) = \mathrm{FALSE} \end{cases} \tag{13}$$

where TRUE and FALSE are matrices with Boolean values of 1 and 0, respectively, which interpret real and fake image pixels identified by the two discriminators.

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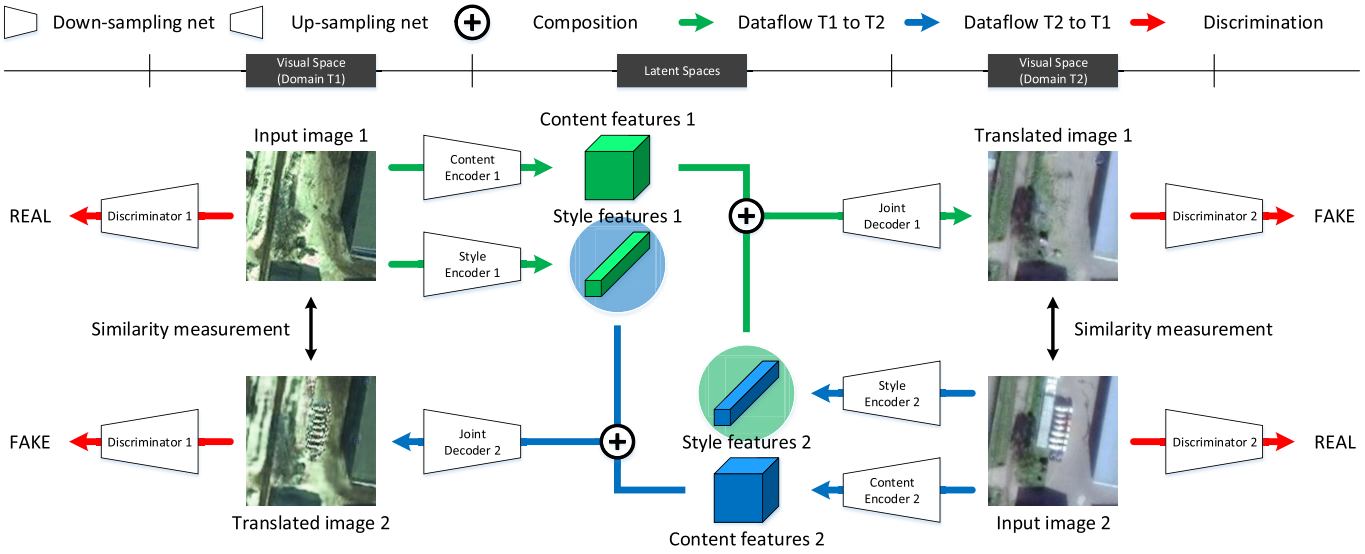


Fig. 2. Framework of our proposed CiDL for change detection in remote sensing images.

B. Loss Function

With the multimodel framework, two novel full objectives are proposed, respectively, for unsupervised and supervised CD using RS images. As the primary parts of these two objectives, several types of terms are listed and interpreted as follows.

1) Adversarial Loss: As the basic loss in GANs, adversarial loss was first proposed by Goodfellow *et al.* [15]. It is set to help the generator fool the discriminator, hindering it from distinguishing fake translated and real original images in the same domain. In this research, we apply two adversarial losses in two domains to facilitate the category-wise cross-domain translation, formulated as

$$\mathcal{L}_{\text{adv}}^{T_1} = \mathbb{E}_{T_1}\left[\log D_1(I_1) + \log(1 - D_1(I_{2\to 1}))\right] \tag{14}$$

$$\mathcal{L}_{\text{adv}}^{T_2} = \mathbb{E}_{T_2}\left[\log D_2(I_2) + \log(1 - D_2(I_{1\to 2}))\right]. \tag{15}$$

By minimizing these losses, all the encoders and decoders try to translate the paired input images to each others domains and make them appear similar to the one another original images. In contrast, by maximizing these losses, the two discriminators try to separate them into the two domains.

2) Visual Reconstruction Loss: In order to guarantee the performance of all the models via pixel-level supervision in the visual spaces, we introduce two visual reconstruction losses, which are derived from the cycle-consistency losses in CycleGAN [43]. They are set here to encourage the two mappings to bring the translated images back to their original ones in the visual spaces, formulated as

$$\mathcal{L}_{\text{vis}}^{T_1} = \mathbb{E}_{T_1}\left[\|I_{1\to 2\to 1} - I_1\|_1\right] \tag{16}$$

$$\mathcal{L}_{\text{vis}}^{T_2} = \mathbb{E}_{T_2}\left[\|I_{2\to 1\to 2} - I_2\|_1\right] \tag{17}$$

where $\|\cdot\|_1$ is the $L1$ distance loss. Minimizing these losses reduces the randomness of all the encoders and decoders, and facilitates the stability of the convergence procedure.

3) Latent Reconstruction Loss: In order to guarantee the performance of all the models via content-and style-level supervision in the latent spaces, we also introduce two content reconstruction losses and two style reconstruction losses, similar to the visual reconstruction losses. The content reconstruction losses are set here to encourage the mappings to maintain the content features of the input images after being translated, formulated as

$$\mathcal{L}_{\text{lat}}^{C_1} = \mathbb{E}_{T_2,T_1}\left[\left\|\text{En}_2^C(I_{1\to 2}) - \text{En}_1^C(I_1)\right\|_1\right] \tag{18}$$

$$\mathcal{L}_{\text{lat}}^{C_2} = E_{T_1,T_2}\left[\left\|\text{En}_1^C(I_{2\to 1}) - \text{En}_2^C(I_2)\right\|_1\right]. \tag{19}$$

While the style reconstruction losses are set here to encourage the mappings to effectively decompose the style features of the input images after being translated, formulated as

$$\mathcal{L}_{\text{lat}}^{S_1} = \mathbb{E}_{T_2,T_1}\left[\left\|\text{En}_1^S(I_{2\to 1}) - \text{En}_1^S(I_1)\right\|_1\right] \tag{20}$$

$$\mathcal{L}_{\text{lat}}^{S_2} = \mathbb{E}_{T_1,T_2}\left[\left\|\text{En}_2^S(I_{1\to 2}) - \text{En}_2^S(I_2)\right\|_1\right]. \tag{21}$$

Minimizing these losses also reduces the randomness of all the encoders and decoders, and facilitates the achievement of the image disentangled representation.

4) Self-Consistency Loss: Due to the powerful feature expression capabilities of deep neural networks, the mappings between two visual domains are essentially stochastic. To stabilize all the models in our framework, we introduce two self-consistency losses, which are derived from the identity losses in CycleGAN [43]. They are set here to ensure that the two image disentangled representations are reversible, formulated as

$$\mathcal{L}_{\text{self}}^{T_1} = \mathbb{E}_{T_1}\left[\|I_{1\to 1} - I_1\|_1\right] \tag{22}$$

$$\mathcal{L}_{\text{self}}^{T_2} = \mathbb{E}_{T_2}\left[\|I_{2\to 2} - I_2\|_1\right]. \tag{23}$$

Minimizing these losses retains the self-consistencies of the images before the decomposition and after the composition in the two domains.

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5) Cross-Consistency Loss: Theoretically, in an ideal situation, paired content features of the registered bitemporal RS images should remain invariant in the unchanged regions, and distinctive differences only exist in the changed regions. Therefore, in the unchanged regions, the appearances of the input and translated images should be visually consistent in the same domain. Along this way, we introduce two cross-consistency losses, which are derived from the cross-consistency losses in DLSF [31]. With the reference, the cross-consistency losses are formulated as

$$\mathcal{L}_{\text{cross}}^{T_1} = \mathbb{E}_{T_1}\left[\|(1-\text{CM}) \otimes (I_{2\to 1} - I_1)\|_2\right] \tag{24}$$

$$\mathcal{L}_{\text{cross}}^{T_2} = \mathbb{E}_{T_2}\left[\|(1-\text{CM}) \otimes (I_{1\to 2} - I_2)\|_2\right] \tag{25}$$

where CM is the binary reference change map that has the same width and height as the input images but only one band, where a value equaling to 1 means that the pixel is part of a changed region, while a value equaling to 0 means that it belongs to an unchanged region. The operation $\otimes$ denotes the element-wise multiplication. $\|\cdot\|_2$ is the $L2$ distance loss. Minimizing these losses facilitates the mappings between two domains.

In general, the full objective of our CiDL is an integration of some of aforementioned loss functions. For unsupervised CD without any reference, the full objective is defined as

$$\begin{aligned}\mathcal{L}_{\text{UCD}} &= \lambda_{\text{adv}}\left(\mathcal{L}_{\text{adv}}^{T_1} + \mathcal{L}_{\text{adv}}^{T_2}\right) + \lambda_{\text{vis}}\left(\mathcal{L}_{\text{vis}}^{T_1} + \mathcal{L}_{\text{vis}}^{T_2}\right) \\ &\quad + \lambda_{\text{lat}}\left(\mathcal{L}_{\text{lat}}^{C_1} + \mathcal{L}_{\text{lat}}^{C_2} + \mathcal{L}_{\text{lat}}^{S_1} + \mathcal{L}_{\text{lat}}^{S_2}\right) + \lambda_{\text{self}}\left(\mathcal{L}_{\text{self}}^{T_1} + \mathcal{L}_{\text{self}}^{T_2}\right). \end{aligned} \tag{26}$$

Alternatively, for supervised CD with some references, the full objective is defined as

$$\begin{aligned}\mathcal{L}_{\text{SCD}} &= \lambda_{\text{adv}}\left(\mathcal{L}_{\text{adv}}^{T_1} + \mathcal{L}_{\text{adv}}^{T_2}\right) + \lambda_{\text{vis}}\left(\mathcal{L}_{\text{vis}}^{T_1} + \mathcal{L}_{\text{vis}}^{T_2}\right) \\ &\quad + \lambda_{\text{lat}}\left(\mathcal{L}_{\text{lat}}^{C_1} + \mathcal{L}_{\text{lat}}^{C_2} + \mathcal{L}_{\text{lat}}^{S_1} + \mathcal{L}_{\text{lat}}^{S_2}\right) \\ &\quad + \lambda_{\text{self}}\left(\mathcal{L}_{\text{self}}^{T_1} + \mathcal{L}_{\text{self}}^{T_2}\right) + \lambda_{\text{cross}}\left(\mathcal{L}_{\text{cross}}^{T_1} + \mathcal{L}_{\text{cross}}^{T_2}\right) \end{aligned} \tag{27}$$

where λ denotes the relative importance of each loss function. Therefore, the main solutions for these two objectives defined above can be expressed as

$$\text{En}_{1,2}^{C*}, \text{En}_{1,2}^{S*}, \text{De}_{1,2}^{*}, D_{1,2}^{*} = \arg \min_{\text{En}_{1,2}^{C}, \text{En}_{1,2}^{S}, \text{De}_{1,2}} \max_{D_{1,2}} \mathcal{L}_{\text{UCD}} \tag{28}$$

$$\text{En}_{1,2}^{C*}, \text{En}_{1,2}^{S*}, \text{De}_{1,2}^{*}, D_{1,2}^{*} = \arg \min_{\text{En}_{1,2}^{C}, \text{En}_{1,2}^{S}, \text{De}_{1,2}} \max_{D_{1,2}} \mathcal{L}_{\text{SCD}}. \tag{29}$$

No matter provided with or without references, all the models are trained simultaneously. When predicting, however, only the well-trained content encoders, style encoders, and joint decoders are utilized for detecting changes.

C. Implementation

1) Network Architecture: Our proposed CiDL is mainly driven by multiple networks, which are two content encoders, two style encoders, two joint decoders, and two domain discriminators. All of the models are CNNs and their architectures are illustrated in Fig. 3. Applying ResNet [50] as the backbone, the content encoder consists of three convolutional blocks and nine residual blocks. This model conducts four-fold down-sampling on the input image patches, producing content feature maps with the size of 64×64. The style encoder consists of five convolutional blocks, one average pooling layer, and one fully connected (FC) layer. This model first conducts sixteen-fold down-sampling on the input image patches, producing style feature maps with the size of 16×16, and then linearly processes them to a one-dimensional vector. In addition, the joint decoder consists of three transposed convolutional blocks, followed by a convolutional layer and a tanh activation layer. Derived from U-Net [51], this model adopts a skip connection technique to composite the given content and style features, producing translated images that have the same size as the original images. In the domain discriminator, a Markovian discrimination strategy [52] is applied to model only the high-frequency structures of paired images. These two models are both PatchGANs that conduct four-fold down-sampling on the paired original and translated images. For all the aforementioned models, the norm layers in their blocks use the instance normalization [53], and their weights and biases are initialized utilizing the strategy of Xavier [54].

2) Training Details: For all the experiments, our primary purpose is to learn well-trained models through the training process that minimizes the full objective L_{UCD} or L_{SCD}. For unsupervised CD, referring to the implementation in MUNIT [49], we set λ_{adv} and λ_{lat} to 1, and set λ_{vis} and λ_{self} to 10 in (26). Alternatively, for supervised CD, referring to the implementation in DLSF [31], we set λ_{cross} to 100 in (27). To stabilize the training process of all the models, we suggest utilizing the least-squares loss [55] specifically for $L_{\text{adv}}^{T_1}$ and $L_{\text{adv}}^{T_2}$ instead of the negative log likelihood loss in basic GANs. In addition, the buffer discrimination strategy [56] is also applied to update the domain discriminators with the historical translated images rather than the latest ones, and the buffer volume is set to 50.

Given the complexity of our hybrid framework, the training process is designed to update all the models alternately. More details about forward and backward propagations in one epoch are presented in Table I.

3) Predicting Details: Although our proposed CiDL learns eight well-trained models, only the two content encoders, two style encoders, and two joint decoders are utilized for predicting. Similar to the training process, the content encoders, En_1^{C*} and En_2^{C*}, are responsible for extracting content features from bitemporal images, while the style encoders, En_1^{S*} and En_2^{S*}, are responsible for extracting style features. Thereafter, the joint decoders, De_1^{*} and De_2^{*}, are responsible for generating translated images by composing different content features with consistent style features. As soon as the paired original and translated images are difficult to separate by the domain discriminators in two domains, our models are able to realize the category-wise cross-domain translation. Finally, we utilize a joint decision maker to perform pixel-level similarity measurements on two couples of original and translated images in

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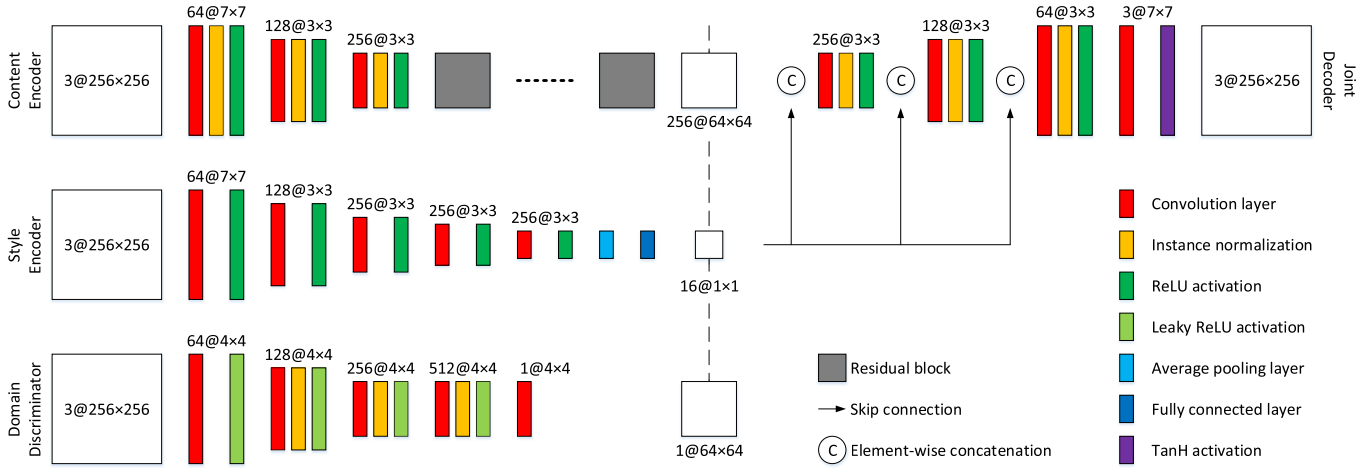


Fig. 3. Architectures of content encoder, style encoder, joint decoder, and domain discriminator.

TABLE I
OVERVIEW OF THE TRAINING PROCESS

Inputs:	Bi-temporal remote sensing images: $\boldsymbol{I_1}$ and $\boldsymbol{I_2}$ Reference change map: $\boldsymbol{CM}$ (for supervised learning)
for $\boldsymbol{epoch \leftarrow 1\ to\ epoch_{max}}$ **do**	
forward 1: $\boldsymbol{I_1, I_2 \rightarrow En^C_{1,2}, En^S_{1,2}, De_{1,2} \Rightarrow I_{1\rightarrow 2}, I_{2\rightarrow 1}}$	
forward 2: $\boldsymbol{I_{1\rightarrow 2}, I_{2\rightarrow 1} \rightarrow En^C_{1,2}, En^S_{1,2}, De_{1,2} \Rightarrow I_{1\rightarrow 2\rightarrow 1}, I_{2\rightarrow 1\rightarrow 2}}$	
forward 3: $\boldsymbol{I_1, I_{2\rightarrow 1} \rightarrow D_1}$	
forward 4: $\boldsymbol{I_2, I_{1\rightarrow 2} \rightarrow D_2}$	
backward 1: update $\boldsymbol{D_1, D_2}$ with $\boldsymbol{\mathcal{L}^{T_1}_{adv}, \mathcal{L}^{T_2}_{adv}}$	
backward 2:	
(for unsupervised learning)	
update $\boldsymbol{En^C_{1,2}, En^S_{1,2}, De_{1,2}}$ with $\boldsymbol{\mathcal{L}_{UCD}}$	
(for supervised learning)	
update $\boldsymbol{En^C_{1,2}, En^S_{1,2}, De_{1,2}}$ with $\boldsymbol{\mathcal{L}_{SCD}, CM}$	
end for	
Outputs:	Well-trained content encoders: En^{C*}_1 and En^{C*}_2 Well-trained style encoders: En^{S*}_1 and En^{S*}_2 Well-trained joint decoders: De^*_1 and De^*_2

two domains, producing a predicted change map, expressed as

$$P(i,j) = \mathbb{E}_{T_1}\left[\|I_{2\rightarrow 1}(i,j) - I_1(i,j)\|_2\right] + \mathbb{E}_{T_2}\left[\|I_{1\rightarrow 2}(i,j) - I_2(i,j)\|_2\right] \quad (30)$$

where P is the predicted change map which reflects the change scores of all pixels. A large value of the pixel located at (i, j) means that the pixel may be part of a changed region, while it close to 0 means that it belongs to an unchanged region. In our research, the final CD result is calculated by the pixel-level binarization of the predicted change map, utilizing a parametric thresholding algorithm.

D. Experiments

In this section, the introduction of data sets, baseline methods, and evaluation metrics are first described in detail. Thereafter, the corresponding experimental setup is provided. Finally, the experimental results of our CiDL and other comparative methods are presented.

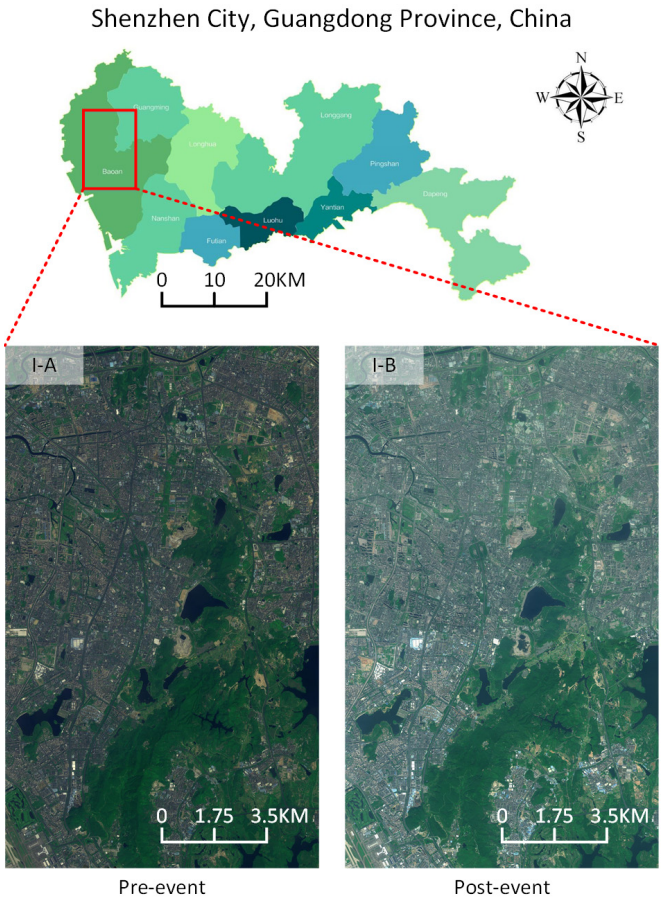


Fig. 4. Overview of data set I with sensor-varying changes.

1) Data Sets Description: The first data set was acquired in Shenzhen City, Guangdong Province, China. It involves extensive sensor-varying changes in dense urban areas [31], covering approximately 182 k. It includes two orthorectified large-scale bitemporal RS images that have different sizes of 5233×8677 and $10\,466 \times 17\,354$, each of them has three bands corresponding to the red (R), green (G), and blue (B), as

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Fig. 5. Representative images in data set II with season-varying changes.

shown in Fig. 4. The pre-event image captured by SPOT 6 in 2014 has a resolution of 2 m per pixel, while the postevent image captured by GeoEye-1 in 2015 has a resolution of 1 m per pixel. The land covers in this data set are classified into six categories: (1) building group; (2) road and highway; (3) tree and vegetation; (4) cultivated land; (5) barren land; and (6) others. According to these classification rules, the binary reference change map is manually drawn, for only assessment in unsupervised learning, or for both training and assessment in supervised learning.

The second data set includes seven pairs of medium scale co-registered RS images with the size of 4275×2700, and four pairs of small scale co-registered ones with the size of 1900×1000, which involve intensive season-varying changes in rural areas [57]. As shown in Fig. 5, several couples of representative images are depicted, and each of them is an RGB optical image. They were obtained by Google Earth (DigitalGlobe) and their resolutions are from 0.03 to 1 meter per pixel. In addition, all the paired RS images have the corresponding reference change maps, which function similarly to those of the first data set.

2) Baseline Methods: In this research, our proposed CiDL is compared with other deep learning models that are designed by current state-of-the-art CD methods. As pioneer works applying deep learning for CD, LSTM [17] and GETNET [18] are the representatives of methods based on original images. PCANet [23] and GAN [24] are the representatives of image algebra-based approaches, and DSCN [26] and CDGAN [30] are the representatives of image transformation-based approaches. DLSF [31], being the previous work of this research, also belongs to approaches based on image transformation. With the above competitors, we perform all the experiments in the supervised learning mode. Moreover, as a supplement to the CD capability of our framework, the unsupervised CiDL is taken into our comparison as well.

3) Evaluation Metrics: To prove the effectiveness and validity of the deep learning-driven CD methods, the following five indices are used to evaluate the accuracy of the experimental results.

Precision (Pre), Recall (Rec), and Overall Accuracy (OA): These three indices are commonly used for assessing the total capabilities of CD models, expressed as

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{31}$$

$$\text{Rec} = \text{TP}/(\text{TP} + \text{FN}) \tag{32}$$

$$\text{OA} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{33}$$

where TP, TN, FP, and FN denote the numbers of true positive, true negative, false positive, and false negative pixels in the predicted change maps, respectively.

4) Kappa Coefficient (KC): This statistical measure represents the consistency between the experimental result and reference, expressed as

$$\text{KC} = \frac{\text{PC} - \text{PE}}{1 - \text{PE}} \tag{34}$$

where PC indicates the true consistency that equals to OA, and PE indicates the theoretical consistency, expressed as

$$\text{PE} = \frac{(\text{TP} + \text{FP})(\text{TP} + \text{FN}) + (\text{FN} + \text{TN})(\text{FP} + \text{TN})}{(\text{TP} + \text{TN} + \text{FP} + \text{FN})^2}. \tag{35}$$

5) F1 Score (F1): This statistical magnitude represents the harmonic average of the precision and recall rates. It is commonly used to assess the performance of deep learning-based models, expressed as

$$\text{F1} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}. \tag{36}$$

For these five evaluation metrics, as the values of Pre, Rec, OA, KC, and F1 become larger, the performance of the corresponding CD method is better.

6) Experimental Setup: Since the paired images in data set I have been orthorectified but without co-registration, it is essential to resample them into the same scale. To prevent the loss of image information, here we suggest resampling the images of high spatial resolution to the ones of low spatial resolution. Along this way, we conduct twofold downsampling on the postevent image to obtain the identical spatial resolution of the pre-event one in data set I. In addition, for both data sets, we perform mean-subtraction normalization on each channel of RS images, from 0 to 255 to -1 to 1, to accelerate the convergence [58] of all the weights and biases in the comparative models.

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TABLE II
EXPERIMENTAL INFORMATION OF CiDL

	Data Pair	Sample Number	Training Time	Testing Rate
Supervised	I-A & I-B	1980	47.05 h	0.31 second/patch
	II.1-A & II.1-B II.1-B & II.1-C II.1-C & II.1-D II.2-A & II.2-B	840	20.19 h	
	II.3-A & II.3-B II.4-A & II.4-B	450	10.96 h	
Unsupervised	I-A & I-B	2640	62.53 h	0.31 second/patch
	II.1-A & II.1-B II.1-B & II.1-C II.1-C & II.1-D II.2-A & II.2-B	1120	26.79 h	
	II.3-A & II.3-B II.4-A & II.4-B	600	14.42 h	

With our two types of data sets, we conducted seven couples of experiments between I-A and I-B, II.1-A and II.1-B, II.1-B and II.1-C, II.1-C and II.1-D, II.2-A and II.2-B, II.3-A and II.3-B, and II.4-A and II.4-B, respectively. In supervised learning, 75% of the paired images are used for training areas, and 25% of those are used for testing areas. In unsupervised learning, all of the paired images are used for both training and testing. As the RS images are of large scale, all the samples utilized in the experiments are small image patches with the identical size of 256×256, and they are randomly cut from the training areas, with certain overlaps and rotations. When testing using a large area, we will first detect changes with small samples, and then splice the predicted change maps together.

For the optimization process, we set 200 epochs and utilize Adaptive Moment Estimation (Adam) [59] as the optimizer, to achieve overall convergence. All the networks are trained from scratch with the learning rate of 10^{-4}. For the first 100 epochs, the learning rate is kept the same and linearly decayed to 0 for the next 100 epochs. The decay rates for the moment estimates are 0.5 and 0.999, and the epsilon is 10^{-8}.

In the present research, our proposed CiDL is implemented in a PyTorch environment, which offers an effective programming interface written in Python. All the experiments are performed on two computers with Intel Core i7, 32-GB RAM, and NVIDIA GTX 1080 GPU. Specific experimental details of our proposed CiDL, including the training times and testing rates, are presented in Table II.

7) Cross-Domain Translation Results: With our two RS data sets, certain representative patch pairs of cross-domain translation results obtained by the supervised and unsupervised CiDL are presented in Fig. 6.

By observing the columns 1 and 2, and 5 and 6, it is notable that, for the input patches of low spatial resolution, our supervised and unsupervised CiDLs produce slightly different cross-domain translation results. There is some degree of data overfitting that occurs in supervised learning, making the later CD results be affected by redundancy to some extent.

By observing the columns 3 and 4, 7 and 8, and 9 and 10, it is notable that, for the input patches of moderate and high spatial resolution, our supervised and unsupervised CiDLs almost produce similar cross-domain translation results, though some land covers have some degree of distortion after the translation from one visual domain to another, as shown in column 4. It is interesting that, despite the absence of references, the unsupervised CiDL sometimes learns a better category-wise translation than the supervised CiDL, as shown in column 3.

In reality, optical RS images generally involve some degree of local spectral variability, that caused by atmospheric effects or image noises, which are widely considered when processing multispectral or hyperspectral RS images [60], [61]. To test the resistance of our models to these issues, we conducted several experiments with our two data sets that are added with different illumination variations and noises. The testing results interpret that, with no matter supervised or unsupervised learning mode, the resistance of our models to the global spectral variability is much stronger than that to the local one. It is because that, the local spectral variability will always mislead the style encoder, hindering it from converging. In general, with the guidance of reference change maps, our supervised CiDL is distinctly more generalized and robust than our unsupervised CiDL.

8) Change Detection Results: With the aforementioned seven bitemporal RS image pairs, the CD results obtained by our CiDL and other competitors are presented in Figs. 7 and 8, where the pixels in white and black indicate the changed and unchanged regions, respectively. The CD results of data set II are generally much better than those of data set I, because the spatial resolution of images in data set II is higher than that in data set I.

For sensor-varying changes, due to the limited capability on feature expression, the deep learning models based on original images, LSTM and GETNET, only give approximate changed regions but with ambiguous outlines, as illustrated in Fig. 7(c) and (d). Although they are insensitive to image noise, a large number of image details are completely overlooked, leading to unsatisfactory CD results. The models based on image algebra, PCANet and GAN, perform better results than original image-based models, but their resistances to image noises are diverse, as illustrated in Fig. 7(e) and (f). It is because that, PCANet is still an end-to-end network, similar to LSTM, GETNET, and CDGAN, but GAN does not. Alternatively, similar diversities appear between DSCN and CDGAN, though they both belong to the models based on image transformation. Compared with other competitors, they can learn more implicit information to improve the resistance to redundancy and the detection ability for subtle changes in RS images, as illustrated in Fig. 7(g) and (h). By integrating the strategies of dual learning and Siamese network, DLSF gives better results than DSCN and GETNET, as illustrated in Fig. 7(i). It is confirmed that the dual learning algorithm benefits the CD tasks. As illustrated in Fig. 7(j), the supervised CiDL, which integrates dual learning algorithm and disentangled representation theory, has displayed the best CD results where the contours of the predicted changed regions are more precise and smoother than the ones by other competitors. It is notable that the unsupervised CiDL is too sensitive to the redundancy and image noise on data set I, and

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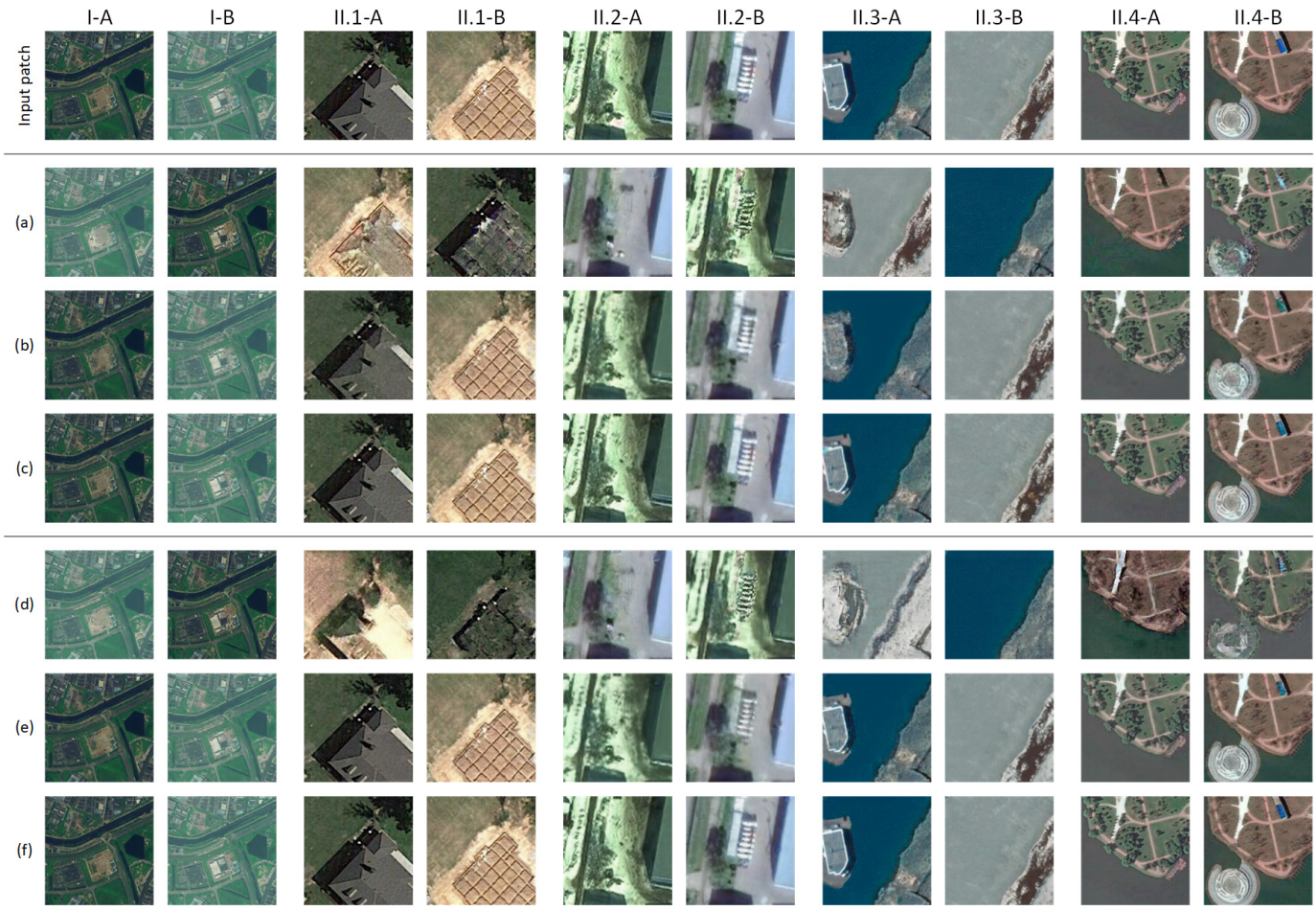


Fig. 6. Representative examples of cross-domain translation results obtained by our CiDL. (a) Translated patch by supervised learning. (b) Reconstructed patch by supervised learning. (c) Self-consistent patch by supervised learning. (d) Translated patch by unsupervised learning. (e) Reconstructed patch by unsupervised learning. (f) Self-consistent patch by unsupervised learning.

there are lots of blurs and uncertainties that are distributed on its CD results, as illustrated in Fig. 7(k).

With the CD results of the comparative models on data set I, the evaluation metrics Pre, Rec, OA, KC, and F1 are computed and summarized in Table III. Compared with LSTM, GETNET, PCANet, GAN, DSCN, CDGAN, and DLSF, our proposed supervised CiDL achieves the highest OA, KC, and F1 values of 0.9301, 0.7823, and 0.8250, respectively, on data set I.

For season-varying changes, depending on different degrees of seasonal variation, the competitors display distinctly diverse CD results on different image pairs.

As shown in Fig. 8(c), due to the weak capability on feature expression, LSTM always gives poor and unstable results that are full of errors and noises. Furthermore, for image pairs with major seasonal variation, it displays incorrect predicted change maps, as depicted in the fourth and fifth blocks in Fig. 8.

As shown in Fig. 8(d) and (e), the results of GETNET and PCANet are sensitive to the degree of seasonal variation. They are able to display stable results for image pairs with minor seasonal variation, but cannot recognize the complete and correct changes for those with major seasonal variation, as depicted in the fourth and fifth blocks in Fig. 8.

As shown in Fig. 8(f), being a hybrid model based on image algebra, GAN performs visibly more robust than PCANet, specifically for image pairs with major seasonal variation. Unlike the end-to-end architecture, the addition of adversarial module can effectively facilitate the model extracting more meaningful domain-invariant features.

As shown in Fig. 8(g), (h), and (i), with the application of Siamese mechanism, three image transformation-based models show different strengths and weaknesses. Among them, DSCN is just a single Siamese network, CDGAN embeds a W-shaped Siamese network in a GAN to build an end-to-end model, and DLSF integrates the dual learning algorithm with two Siamese networks to build a hybrid model. In comparison, DSCN gives the worst results, where numerous predicted change pixels are distributed discretely. Unconventionally, it sometimes displays completely incorrect change maps, as depicted in the third and sixth blocks in Fig. 8. Remarkably, CDGAN is able to perform much better, even producing a perfect change map specifically for image pair II.1-A and II.1-B. It is confirmed that end-to-end models can still work with some advanced modules. However, for image pairs with major seasonal variation, it always fails to capture subtle changes, such as vehicles

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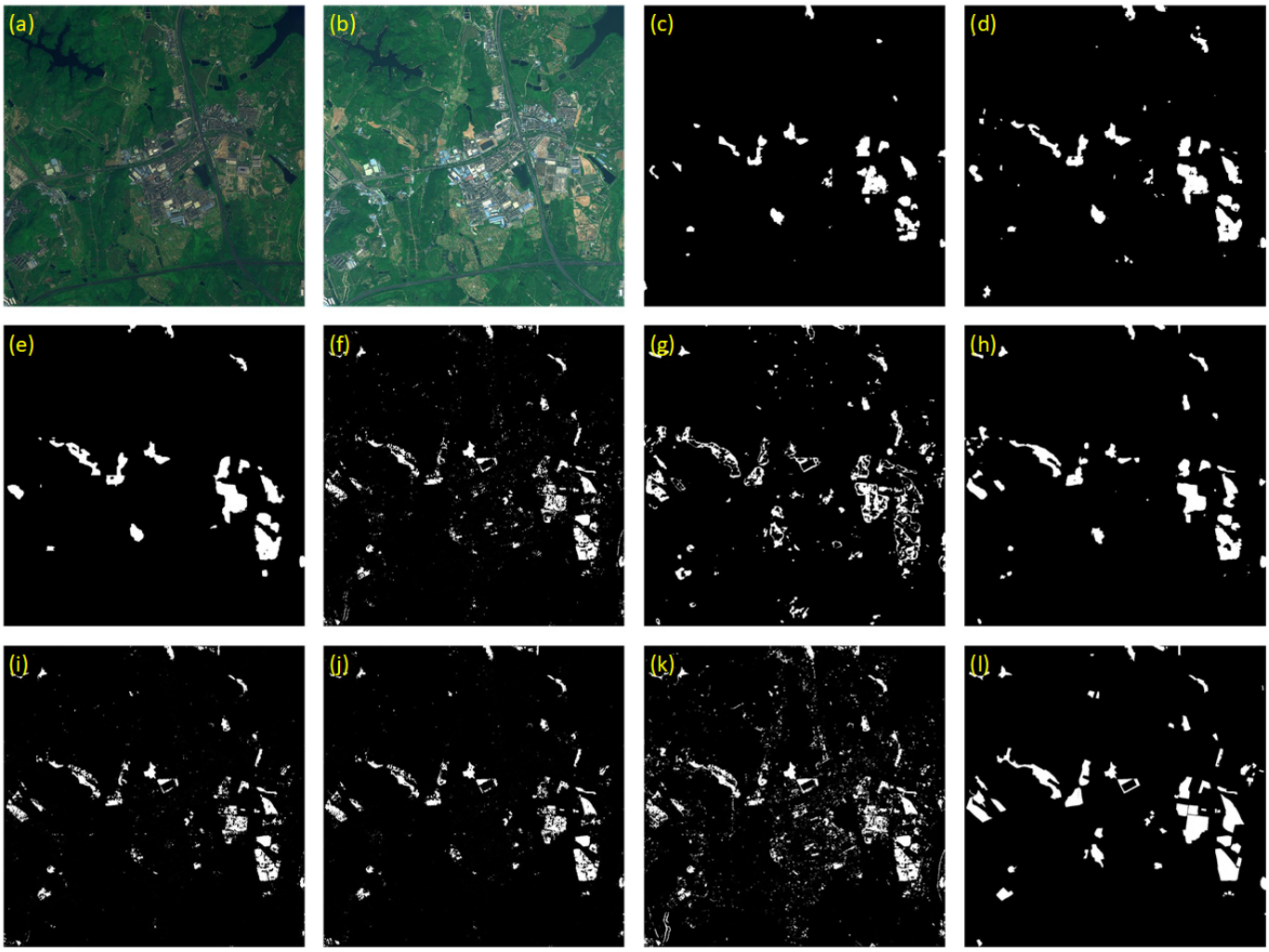


Fig. 7. Change detection results of the comparative methods on data set I. (a) Pre-event image. (b) Postevent image. (c) LSTM. (d) GETNET. (e) PCANet. (f) GAN. (g) DSCN. (h) CDGAN. (i) DLSF. (j) Supervised CiDL. (k) Unsupervised CiDL. (l) Ground truth.

TABLE III
QUANTITATIVE RESULTS OF THE COMPARATIVE METHODS ON DATA SET I. THE BEST VALUES ARE IN BOLD

Data Pair	Metric	LSTM	GETNET	PCANet	GAN	DSCN	CDGAN	DLSF	CiDL Sup.	CiDL Uns.
I-A & I-B	Pre	0.8396	0.8486	0.9193	0.6678	0.6619	**0.9202**	0.7262	0.7277	0.4476
	Rec	0.5381	0.6273	0.6670	0.8793	0.8375	0.6959	0.9282	**0.9524**	0.9129
	OA	0.7961	0.8350	0.9085	0.8407	0.8263	0.9250	0.8986	**0.9301**	0.8196
	KC	0.5207	0.6079	0.7176	0.6434	0.6118	0.7478	0.7465	**0.7823**	0.5012
	F1	0.6559	0.7214	0.7731	0.7591	0.7394	0.7925	0.8149	**0.8250**	0.6007

and roads, as depicted in the fourth and fifth blocks in Fig. 8. As the previous work of this research, DLSF appears to be too complex for two parallel data streams. Although its results seem to be satisfactory, there is high possibility that data overfitting occurs when training the model, making it vulnerable to spectral variability, as depicted in the fifth block in Fig. 8.

As shown in Fig. 8(j), the supervised CiDL always performs effectively and precisely for image pairs with no matter minor or major seasonal variation. Applying dual learning algorithm and disentangled representation theory, it can not only capture almost all subtle changes, but also alleviate the negative influences of redundancy and image noise. In addition, as shown in Fig. 8(k), even without references, the unsupervised CiDL can still display satisfactory results, though they are not as accurate and complete as those by the supervised CiDL. It is confirmed that our multimodel framework is effective and robust for CD tasks. Interestingly, the unsupervised CiDL occasionally gives more meaningful changes than the supervised one, as depicted in the first block in Fig. 8. This may provide certain inspiration for recognition tasks on change categories.

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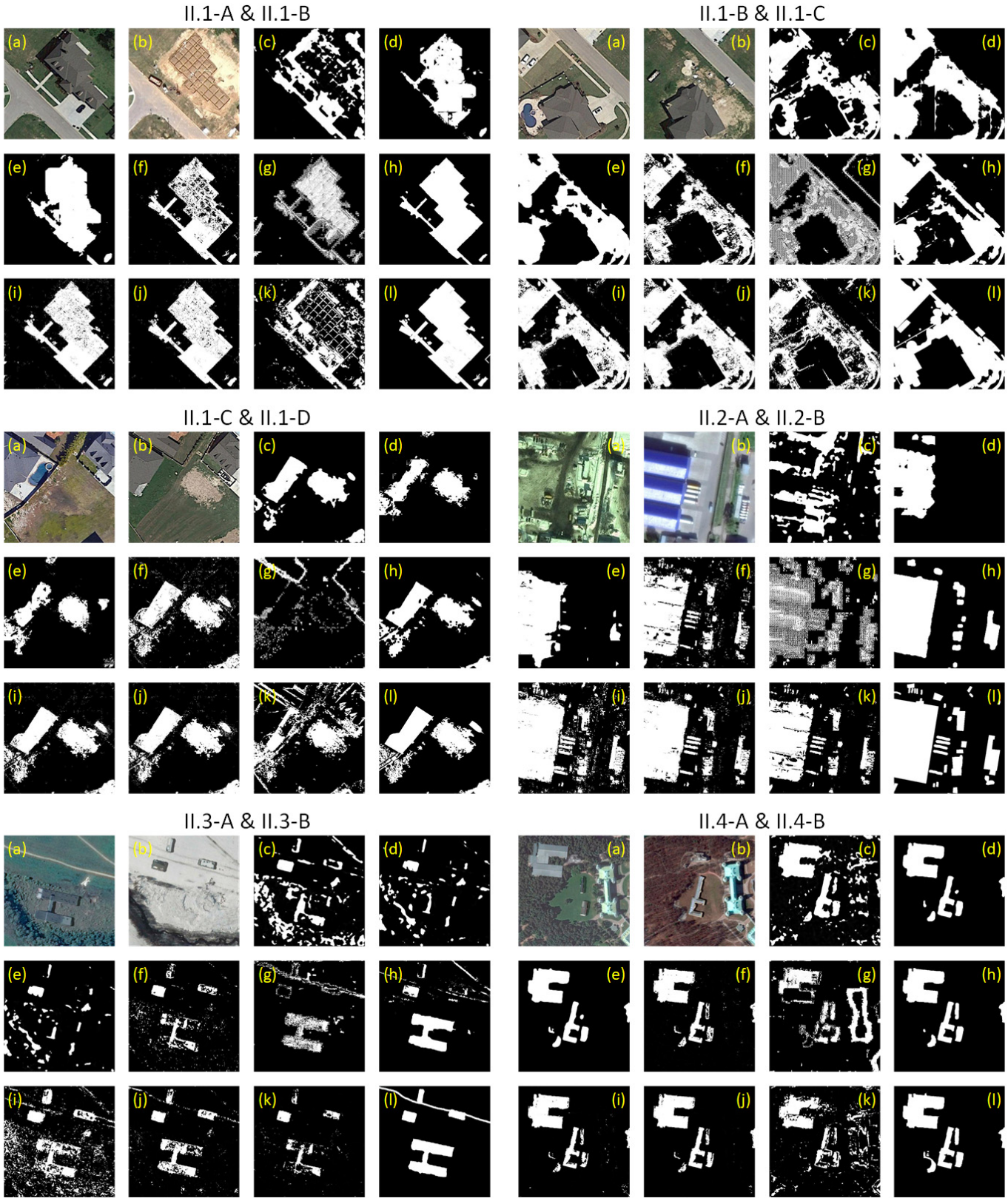


Fig. 8. Change detection results of the comparative methods on data set II. (a) Pre-event image. (b) Postevent image. (c) LSTM. (d) GETNET. (e) PCANet. (f) GAN. (g) DSCN. (h) CDGAN. (i) DLSF. (j) Supervised CiDL. (k) Unsupervised CiDL. (l) Ground truth.

With the CD results of the comparative models on data set II, the evaluation metrics Pre, Rec, OA, KC, and F1 are computed and summarized in Table IV. Compared with LSTM, GETNET, PCANet, GAN, DSCN, CDGAN, and DLSF, our proposed supervised CiDL achieves competitive OA, KC, and F1 values, on data set II.

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TABLE IV
QUANTITATIVE RESULTS OF THE COMPARATIVE METHODS ON DATA SET II. THE BEST VALUES ARE IN BOLD. (-) INDICATES NEGATIVE VALUE

Data Pair	Metric	LSTM	GETNET	PCANet	GAN	DSCN	CDGAN	DLSF	CiDL Sup.	CiDL Uns.
II.1-A & II.1-B	Pre	0.5719	0.6570	0.6124	0.8992	0.9034	0.9794	0.9493	**0.9812**	0.8265
	Rec	0.3540	0.7705	0.8058	0.6317	0.6181	**0.8351**	0.7394	0.6766	0.5625
	OA	0.6805	0.8981	0.9212	0.9069	0.8863	**0.9772**	0.9225	0.9697	0.8834
	KC	0.2311	0.6479	0.6516	0.6874	0.6649	**0.8887**	0.7820	0.7851	0.6017
	F1	0.4373	0.7092	0.6959	0.7421	0.7340	**0.9015**	0.8313	0.8009	0.6694
II.1-B & II.1-C	Pre	0.7367	0.8216	0.7838	0.8803	0.8378	0.7754	0.9151	**0.9287**	0.8225
	Rec	0.5580	0.8806	**0.8893**	0.5913	0.4619	0.8789	0.7979	0.6893	0.5478
	OA	0.7921	0.8786	0.8918	0.8706	0.7818	0.9087	0.8946	**0.9198**	0.8102
	KC	0.4935	0.7483	0.7536	0.6284	0.4627	0.7626	**0.7711**	0.7430	0.5335
	F1	0.6350	0.8501	0.8332	0.7074	0.5955	0.8239	**0.8525**	0.7913	0.6576
II.1-C & II.1-D	Pre	0.7816	0.7972	0.8161	0.6944	0.3491	**0.8440**	0.7035	0.7630	0.4884
	Rec	0.6974	0.5793	0.6461	0.9267	0.2092	0.6685	0.9394	**0.9614**	0.7958
	OA	0.8834	0.8911	0.9063	0.9134	0.4034	0.9089	0.9146	**0.9587**	0.8271
	KC	0.6625	0.6077	0.6658	0.7405	-	0.6915	0.7513	**0.8272**	0.5026
	F1	0.7371	0.6710	0.7212	0.7939	0.2616	0.7461	0.8045	**0.8508**	0.6053
II.2-A & II.2-B	Pre	0.3765	0.7841	0.7125	0.6444	0.8506	**0.8735**	0.5826	0.6621	0.5963
	Rec	0.2271	0.2826	0.3129	0.9205	0.5395	0.5602	0.9310	**0.9615**	0.8731
	OA	0.4556	0.6443	0.6705	0.9041	0.8094	0.8766	0.9158	**0.9482**	0.8633
	KC	-	0.2339	0.2492	0.7006	0.5368	0.6105	0.6703	**0.7559**	0.6234
	F1	0.2833	0.4155	0.4348	0.7581	0.6602	0.6826	0.7167	**0.7842**	0.7086
II.3-A & II.3-B	Pre	0.3119	0.5034	0.5131	0.8124	0.8512	**0.8717**	0.4820	0.6573	0.6196
	Rec	0.3840	0.1989	0.2056	0.4678	0.4770	0.6508	0.8442	**0.8777**	0.4669
	OA	0.5711	0.4123	0.4328	0.7815	0.8176	0.9365	0.8575	**0.9473**	0.7697
	KC	0.0306	-	-	0.4587	0.5043	0.7098	0.5341	**0.7229**	0.3837
	F1	0.3442	0.2851	0.2936	0.5937	0.6114	0.7452	0.6136	**0.7517**	0.5325
II.4-A & II.4-B	Pre	0.5839	0.9454	0.9738	0.9362	0.4279	0.9211	0.9635	**0.9864**	0.6680
	Rec	0.6712	0.6528	0.6848	0.5944	0.7107	**0.7152**	0.5884	0.6581	0.5417
	OA	0.7987	0.9174	0.9332	0.9115	0.7003	0.9455	0.9291	**0.9630**	0.8554
	KC	0.4879	0.7239	0.7653	0.6774	0.3328	**0.7741**	0.6925	0.7701	0.5113
	F1	0.6245	0.7723	0.8041	0.7271	0.5342	**0.8052**	0.7306	0.7895	0.5983

IV. DISCUSSION

For deep learning-driven CD methods utilizing multimodel frameworks, the experimental performance is generally dominated by two decisive factors, which are the model architecture and full objective. In this section, the discussions on our design choices behind these two factors are performed.

A. Design of Model Architectures

Deep learning networks have considerably diverse structures for different image processing tasks. Networks with large and complex structures generally have strong capabilities in feature extraction and representation, but they may induce data overfitting to some extent. On the contrary, networks with small and simple structures are of high efficiency and generalization, but their limited expression may reduce the utilization of image information. Therefore, the optimal design of model architectures is essential for better experimental performance. With regard to the CD using RS images, we specifically design all the models with quantitative experiments to adapt the training samples and purposes. The designs of the structures of the content and style encoders in our CiDL are analyzed in detail as follows.

B. Content Encoder

Taking ResNet as the backbone, we design this encoder consists of two down-sampling layers and nine residual blocks, to process the input patches into middle-level content features. As optical RS images provide considerable detailed land cover information, the changes in bitemporal RS images are generally complicated. Thus, the down-sampling processes are essential to facilitate training the networks to extract implicit high-level features. Moreover, the residual blocks are also essential as the advanced units of the network, which determine the capability volumes of the models. To design an optimal content encoder, we took data set I as the study areas, and conducted several experiments utilizing diverse content encoders and the same style encoder. The primary differences of these comparative content encoders are the numbers of down-sampling processes and residual blocks. With the testing results obtained by these different model combinations, the OA values of them are computed and summarized in Fig. 9(a).

From the perspective of down-sampling process, the testing results will be improved as the sampling number increases. The testing results are bad with only one down-sampling layer even this model consists of enough residual blocks. Conversely, the content encoder equipped with more than two down-sampling layers does not significantly improve the testing results. That is the reason why most cross-domain translation models choose to conduct only two down-sampling processes.

From the perspective of residual block, the testing result can be distinctly improved as the residual block number increases, with more than one down-sampling process. It shows a

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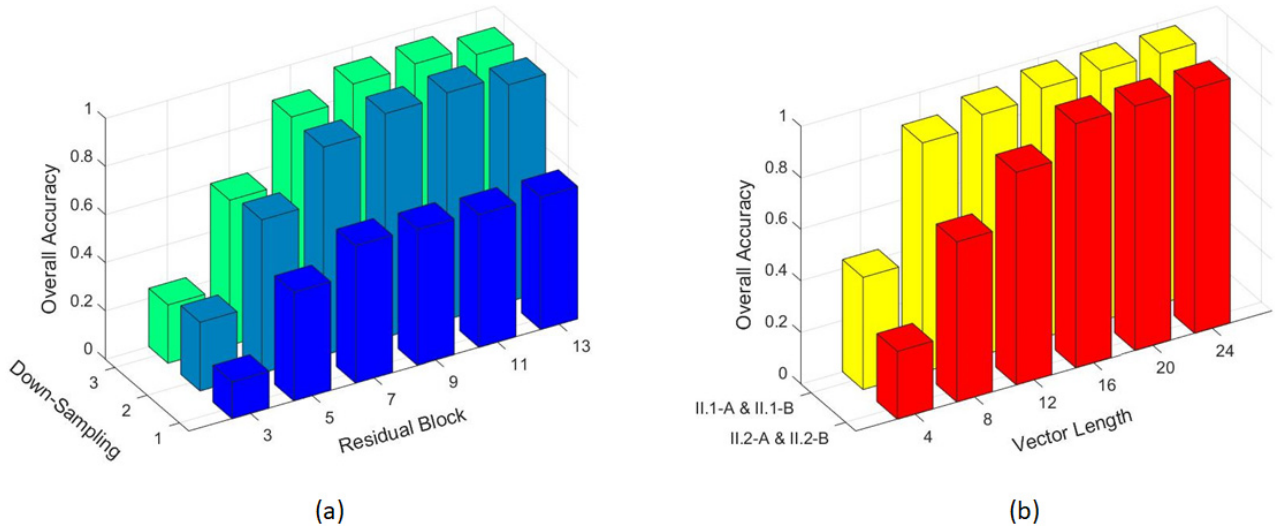


Fig. 9. Comparison of different model architectures based on the supervised CiDL. (a) For content encoders. (b) For style encoders.

TABLE V
ABLATION STUDIES OF THE FULL OBJECTIVE FOR OUR SEVEN EXPERIMENTS

Loss Combinations	I-A&I-B	II.1-A&II.1-B	II.1-B&II.1-C	II.1-C&II.1-D	II.2-A&II.2-B	II.3-A&II.3-B	II.4-A&II.4-B
adv	0.3813	0.4122	0.3771	0.3956	0.2939	0.2630	0.3867
adv+vis	0.5280	0.5671	0.5049	0.5423	0.4906	0.4375	0.5544
adv+vis+lat	0.7833	0.8586	0.7572	0.8005	0.8289	0.7255	0.8301
adv+vis+lat+self	0.8196	0.8834	0.8102	0.8271	0.8633	0.7697	0.8554
adv+vis+lat+cross	0.9145	0.9502	0.9077	0.9336	0.9190	0.8843	0.9214
adv+vis+lat+self+cross	0.9301	0.9697	0.9198	0.9587	0.9482	0.9473	0.9630

significant peak at 11 residual blocks, as illustrated by the green and cyan bars in Fig. 9(a). In order to ensure the training efficiency of the models and avoid data overfitting, we followed the laws revealed by the statistics and then designed the content encoder which is equipped with two down-sampling processes and nine residual blocks.

C. Style Encoder

The primary purpose of this style encoder is to extract high-level style features from the input patches, by processing them into tiny one-dimensional vectors. The length of these vectors determines the capability volumes on recording style features. Longer vectors are able to record more style information, but also increase the complexities of training processes, especially for the backward propagation procedure. To design an optimal style encoder, we took image pairs II.1-A and II.1-B and II.2-A and II.2-B as the study areas, and conducted several experiments utilizing the same content encoder and diverse style encoders. The primary difference of these comparative style encoders is their late FC layers, which process the style feature maps into vectors with different lengths. With the testing results obtained by these different model combinations, the OA values of them are computed and summarized in Fig. 9(b).

As can be seen, for the image pair with minor seasonal variation, such as II.1-A and II.1-B, the style vector with the size of $8 \times 1 \times 1$ has sufficient capability to record the category-wise style information. Nevertheless, for the image pair with major seasonal variation, such as II.2-A and II.2-B, the style vector is not powerful enough to give satisfactory testing results, unless it is augmented to the size of $16 \times 1 \times 1$, as illustrated by the yellow and red bars in Fig. 9(b). Considering the complexities and diversities of RS images, we followed the laws revealed by the statistics and then designed the style encoder that processes the input patches to vectors with larger sizes, though most conventional cross-domain translation models, such as DRIT and MUNIT, still choose to utilize smaller style vectors.

D. Design of Loss Functions

As the representative of training goal, the loss functions determine the qualities of the well-trained models. To validate the effectiveness and uniqueness of our loss functions, we conduct several ablation studies separately for all the experiments using the supervised CiDL. With the testing results guided by six different loss combinations, the OA values of them are computed and summarized in Table V. Detailed interpretations about the effects of all loss functions are outlined in the following.

It is noteworthy that the fourth combination represents our unsupervised CiDL, and the last one represents our supervised CiDL. As can be seen, when the full objectives become more complex, the testing OAs are continuously growing. It is confirmed that complex loss combinations are of effectiveness and generalization for testing samples.

By comparing the first, second, and third rows, it is evident that, without latent content and style losses, our framework is

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unable to implement the purposeful cross-domain translations, leading to completely meaningless CD results. Significant differences in OAs indicate that these two types of constraints are vital for CD when using our proposed CiDL.

By comparing the third and fifth, or fourth and sixth rows, it is evident that the addition of cross-consistent loss can prominently improve the performance. The distinct increases in AOs indicate that this constraint can effectively facilitate the two Y-Nets to better learn the category-wise cross-domain translation between bitemporal RS images.

By comparing the third and fourth, or fifth and sixth rows, it is evident that the addition of self-consistent loss can slightly improve the performance. In practice, this constraint is widely adopted in cross-domain translation multimodel frameworks, to make the training process rid of model oscillations.

V. Conclusion

In this article, we integrated dual learning algorithm and disentangled representation theory, and developed a novel image transformation-based method, namely, CiDL, for CD using RS images. With the designed content and style encoders, we successfully decomposed the input bitemporal images by extracting their latent content and style features. Thereafter, the joint decoders successfully achieve the composition and reconstruction of these images from the latent spaces to visual ones. With the two opposite data streams and two Y-Nets, we successfully learned a category-wise cross-domain translation, which can effectively suppress the differences in paired unchanged regions while highlighting those in paired changed regions. Compared with other state-of-the-art deep learning models in supervised learning, the experiments on two typical CD data sets validated the effectiveness and competitiveness of our CiDL. In addition, even without references, our models can still yield satisfactory CD performance.

Nevertheless, our proposed method still involves two major limitations. Firstly, our cross-domain translation is sensitive to irregular tilts of tall constructions, which are commonly visible in RS images with high spatial resolution and generally caused by diverse view angles of different satellites. This problem can mislead the content encoders, leading to model collapse during the training process. Secondly, in unsupervised learning mode, our multimodel framework occasionally gives poor translated images, where the same type of land covers located at different positions display significantly diverse appearances. According to rigorous analyses, we believe that our framework may learn multiple types of style features in the discrimination buffer and combine them stochastically with the same type of content features. Therefore, in future studies, we plan to take a closer look at the mechanism of disentangled representation, then explore more optimal constraints in the latent spaces to further improve the performance of our proposed CiDL.

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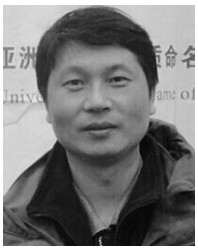
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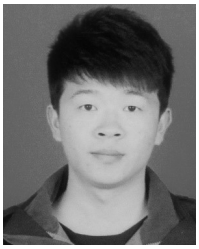
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